2023 SSMO Tiebreaker Round

SMO Team

Tiebreaker Round Problem 1: From the phrase "Summer Solstice" (treating uppercase, lowercase, and the space as distinct, and using each character at most as many times as it appears), how many 4-character strings can be formed such that the first character is a capital letter and the second and third are not spaces?

Tiebreaker Round Problem 2: Let $P(x) = x^3 + 3mx^2 + 3nx + (m+n)$ be a real polynomial with positive real roots p, q, r (not necessarily distinct). Suppose that

$$(p+1)^3 + (q+1)^3 + (r+1)^3 + 3P(-1) = 0.$$

The value of $P(1)$ can be expressed as $a + b\sqrt{c}$, where a, b, and c are integers and c is not divisible by the square of any prime. Find $a + b + c$.

Tiebreaker Round Problem 3: For $n \geq 4$, let a_n be the maximum possible value of $P(n+1)$ given that $P(x)$ is a monic polynomial of degree n that satisfies $P(i) \in \{1, 2, 3, \ldots, n\}$ for $1 \leq i \leq n$. The value of $\sum_{n=4}^{\infty} \frac{a_n - n!}{3^n}$ can be expressed as $\frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$.